MSG No. 84.51 $P4_2/m$ [Type I, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,4,5,8]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,3,6,7]`

Table 2: Wyckoff site: `2b`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,4,5,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,6,7]`

Table 3: Wyckoff site: `2c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,8]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,6,7]`

Table 4: Wyckoff site: `2d`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,5,8]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,6,7]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,6,7]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,5,8]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,4,6,7]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,5,8]`

Table 7: Wyckoff site: `4g`, site symmetry: `2..`

No.	position	mapping
1	$[0, 0, z]$	`[1,4]`
2	$\left[0, 0, z + \frac{1}{2}\right]$	`[2,3]`

continued ...

Table 7

No.	position	mapping
3	$[0,\ 0,\ -z]$	`[5,8]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[6,7]`

Table 8: Wyckoff site: `4h`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5,8]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[6,7]`

Table 9: Wyckoff site: `4i`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5,8]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[6,7]`

Table 10: Wyckoff site: `4j`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8]`
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[2,7]`
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[3,6]`
4	$[-x,\ -y,\ 0]$	`[4,5]`

Table 11: Wyckoff site: `8k`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	`[6]`
7	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	`[7]`
8	$[x,\ y,\ -z]$	`[8]`